INCH-POUND
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22 April 2026
SUPERSEDING
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MILITARY SPECIFICATION

CANTEEN, WATER, COLLAPSIBLE, 2-QUART CAPACITY

This document is approved for use by all Departments and Agencies of the Department of Defense (DoD).

1. SCOPE

1.1 Scope. This specification covers a 2-quart capacity collapsible, plastic water canteen with or without a cover.

2. APPLICABLE PUBLICATIONS.

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government publications.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

Comments, suggestions, or questions on this document should be sent to trpsptctstandardiz@dla.mil. Since contact information can change, verify the currency of the address information using ASSIST https://assist.dla.mil.

AMSC N/A FSC 8465

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DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-32800 - Cap, Non-Nuclear, Biological, Or Chemical (NBC), Water Canteen, Field, 1 Quart And 2 Quart Canteens
MIL-C-43689 - Cover, Water Canteen, 2 Quart, Collapsible (with Pile Lining)
MIL-C-51278 - Cap, Water Canteen, Field, 1 Quart and 2 Quart Canteens

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract.

DRAWINGS

U.S. ARMY, DEVCOM - SOLDIER CENTER

8-2-228 - Canteen, Water, Collapsible, 2-Quart Capacity; Assembly

(Copies of drawings are available from the U.S. Army, DEVCOM Soldier Center, ATTN: PESD-SCB 10 General Greene Avenue, Natick, MA 01760-5019.)

OTHER GOVERNMENT DOCUMENTS AND PUBLICATIONS

CODE OF FEDERAL REGULATIONS

Title 21 Code of Federal Regulations (CRF) Food and Drugs.

(Copies of these documents are available online at https://www.ecfr.gov.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ/ANSI Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at https://asq.org.)

ASTM INTERNATIONAL

ASTM D638 - Standard Test Method for Tensile Properties of Plastics
ASTM D1505 - Standard Test Method for Density of Plastics by the Density Gradient Technique.

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ASTM D1525 - Standard Test Method for Vicat Softening Temperature of Plastics
ASTM D4976 - Standard Specification for Polyethylene Plastic Molding and Extrusion Materials

(Copies of these documents are available online at https://www.astm.org.)

SAE INTERNATIONAL

SAE-AMS-STD-595 - Colors Used in Government Procurement

(This standard and all related color media may be obtained from https://www.sae.org or SAE International Customer Service, 400 Commonwealth Drive, Warrendale, PA 15096.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.1, 4.3.1 6.2, and 6.3).

3.2 Materials.

3.2.1 Resin purity. The ethylene-vinyl acetate (EVA) copolymer used in the manufacture of the canteen bladder and the polyethylene used in the manufacture of the canteen; plus, other substances used in the fabrication of the plastic shall conform to the requirements of 21 CRF -Part 177 Indirect Food Additives: Polymer.

3.2.1.1 Plastic canteen bladder. The plastic canteen bladder shall be made from an EVA copolymer and shall contain no plasticizers. The vinyl acetate content shall be 18.5 (± 2) percent. The copolymer resin shall meet the requirements of Table I when tested as specified in 4.4.1.

TABLE I. Physical characteristics of EVA resin.

Characteristic	Requirement	
	Minimum	Maximum
Density, g/cm^3 at 23.4 ℃	0.935	0.943
Vicat softening point, ℃	57	-
Tensile strength, pounds per square inch, psi	2000	-
Elongation, percent	650	-

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3.2.1.2 Plastic canteen strap. The plastic strap of the canteen shall be made of low density polyethylene conforming to Group 1, Class 1 Grade 2 of ASTM D4976. Clean, unfumed material in the form of imperfect parts, spruces, runners or other scraps of the same composition as the virgin material and produced in the molding or finishing operation, may be reground and mixed with the specified virgin material. When reground is mixed with virgin material, the reground shall not exceed a level of 25 percent by weight of the blended mixture.

3.2.1.3 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Design and construction. The design and construction of the canteen shall be as specified herein and as shown on the Drawing listed in 2.1. When the canteen is to be furnished with or without cover (see 6.2), the cover shall be in accordance outlined in this document. The canteen capacity shall be not less than 1905 mL. of water when tested as specified in 4.3.3. The color of the bladder, and strap shall approximate color No. 34087 olive drab lusterless of AMS-STD-595. The strap shall match the shade of the bladder body and neck.

3.3.1 Bladder. The bladder shall be blow molded from the material specified in 3.2.1.1 by a process in which the bladder is blown and molded in one (1) piece and not as a result of a sheet forming operation. The neck may be molded separately and fused to the bladder in the blown molding operation. The lip of the neck which forms and sealing surface against the cap on assembly shall be flat, smooth and free from sink marks and depressions. The canteen neck shall be threaded with buttress type threads for a minimum of two full turns. The threads shall be fully formed and free from flash. The strap retention groove shall not be malformed, nor have a rough or feathered edge which may cause binding. There shall be no interference with the engagement of the cap to its fully seated position. The wall thickness at any point on the bladder shall be not less than 0.012 inch nor more than 0.056 inch, except as shown on Drawing 8-2-228 and at the parting line where the maximum thickness shall be 0.085 inch. The bladder shall be blow molded in such a manner that the material will be distributed to conform to the shape and general overall design as shown on Drawing 8-2-228. The bladder outer surface, except the identification area, shall have an embossed pebble grain finish to match the guide sample (see 6.3). All parting lines shall be free from flash except that flash extending not more than 1/32 inch at the bottom pinch-off area of the bladder shall be permitted. The grain shall be produced by the finish on the mod cavity and not by a protective coating or lacquer or other materials. The bladder weight shall be not greater than 110 grams and shall not be less than 60 grams when tested as specified in Table V.

3.3.2 Strap. The strap shall be an injection molded from the material specified in 3.2.1.2. Any die parting lines which may appear on the outside surface of the strap shall appear as lines only. No flash shall appear at the parting line. All gates shall be trimmed flush with the molding surface. The exterior surfaces of both sides of the strap shall have a lusterless finish throughout and shall be of the uniform olive drab color specified in 3.3. The strap shall be free of dirt, dust, grease and foreign matter. The strap finish shall be produced from a die which has cavity

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surfaces finished in vapor blast, water hone or similar satin finish and not by the application of any protective coasting, lacquers or other materials.

3.3.3 Cap. The cap shall be in accordance as specified in MIL-DTL-32800.

3.4 Performance.

3.4.1 Porosity. The canteen shall not leak air when tested as specified in Table V.

3.4.2 Leak resistance. The canteen shall show no leakage of water when tested as specified in 4.5 and Table V.

3.4.3 Drop resistance. The canteen shall show on evidence of leakage of water or damage to the bladder, when tested as specified in 4.5 and Table V.

3.5 Marking. One side of the bladder shall be permanently marked by an embossed molded in process, with US, year of manufacture, and contractors name marking shall be clearly legible, and shall not impair the strength of the wall to which it is applied.

3.6 Toxicity. Unless otherwise specified in the contract or procurement documents, the finished canteen shall not present a health hazard and shall show compatibility with prolonged, direct skin contact when tested as specified in 4.6.

3.7 Workmanship. The finished canteen shall be uniform in quality and shall conform to the quality of product established by this specification. The occurrence of defects, as specified in 4.5.1 and Table IV, shall not exceed the criteria specified in the contract or purchase order.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3.1)
b. Conformance inspection (see 4.3.2)

4.2 Inspection conditions. Unless otherwise specified, excluded, amended, modified or qualified in this specification or applicable procurement documents (see 6.2), all inspections shall be performed in accordance with all the requirements of referenced documents.

4.3 Types of inspections.

4.3.1 First article inspection. The first article, submitted in accordance with 3.1 shall be inspected for appearance, color, finished measurements listed in Table IV, examination for defects in 4.5 and tested for the characteristics as specified in 4.5 through 4.5.1.

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4.3.2 Conformance inspection. Conformance inspection shall include the examination of 4.4 and the tests of 4.5 through 4.5.1 as applicable. Sampling for inspection shall be performed in accordance with ASQ/ANSI Z1.4 and with acceptable quality limits (AQLs) as specified in the contract and/or procurement documents, except where otherwise indicated (see 6.2).

4.3.3 Component and material inspection. In accordance with 4.1 components and material shall be inspected in accordance with all the requirements of referenced specifications, drawing and standard unless otherwise excluded amended, modified or qualified in this specification or applicable purchase document. In addition, the physical characteristics of the EVA copolymer resin specified in 3.2.1.1 shall be tested in accordance with Table II. The sample unit shall be a 12 inch by 12 inch slab, which may be unpigmented, having a uniform thickness of 75 (±) 10 mils, and shall be molded from the same EVA material used to manufacture the canteens. The sample size (number of sample; units to be tested) shall be two (2). There shall be no failure of any sample unit to meet the requirements as specified. All test reports shall contain the individual values utilized in expressing the final result. (See Table II).

TABLE II. Component testing.

Characteristic	Requirement paragraph	Test method	No. or determinations per sample unit	Results reported to nearest
EVA resin: Density	3.2.1.1	ASTM D1505	Average of 3	0.001 g/cm^3
Vicat softening	3.2.1.1	ASTM D1525 1/, 2/	Average of 5	1 ^{oC}
Tensile strength (psi) at yield	3.2.1.1	ASTM D638 1/	Average of 6 (3 in each direction)	5 psi
Elongation (percent) at yield	3.2.1.1	ASTM D638 1/	Average of 6 (3 in each direction)	5 percent

1/ Specimens shall be compression molded.
2/ Rate A shall be used for rate of temperature increase, and retesting provisions shall not be applicable.

4.4 Components and material testing.

4.4.1 Material certification. Components and material listed below may be accepted on the basis of a certificate of compliance for requirements specified in applicable paragraphs of this specification. (See Table III).

TABLE III. Components and materials.

Component	Characteristic	Requirement paragraph
Plastic material	Purity	3.2.1
	Absence of plasticizers	3.2.1.1
	Vinyl acetate content	3.2.1.1

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4.5 Inspection of the end item.

4.5.1 Visual examination. The canteen shall be examined for the defects listed in Table IV.

TABLE IV. End item visual defects.

Examine	Defect	Classification	
		Major	Minor
Quality and finish of canteen, neck, and strap	Color not as specified.	101	
	Exterior surfaces of both sides of the strap not a lusterless finish.		201
	Any bubble, sharp crease, wrinkle, blister or gouge	102 * 1/	*1/ 202
	Any crazing, pit, chipping.	103	
	Outside surfaces of neck including threaded portion not smooth	104	
	Finish produced by application of coating, lacquer, or other organic material.	105	
	Any cut, tear, hole, burn, break, or crack in canteen.	106	
	Weld mark, sink mark, shrink mark, dulling of surfaces, roughness, or abrasion in plastic.	107 *1/	203 *1/
	Large permanent indentation, dimpling, or permanent set.	108 *1/	204 *1/
	Canteen has objectionable or vinegar like order.	109	
Design and construction	Varies from specified design and construction shown in specified drawing.	110 * 1/	205 *1/
	Not shape or contour specified.	111 * 1/	206 * 1/
	Seam construction not as specified.	112	
	Taper in neck in excess of that permitted by drawing.	113 * 1	207* 1
Workmanship and assembly (general)	Any open seam.	114	
	Any patch, or repair of plastic material.	115	
	Malformed, distorted or out of round.	116 *1/	208 *1/
	Any foreign matter or material not readily removeable on the outside of the canteen that may cause damage to the material or have an injurious effect on the user.	117	
	Any excessive lump on the inside or outside of the canteen, and of the same material as the canteen, not readily removable.	118 * 1/ 207	

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TABLE IV. Defects. - Continued

Examine	Defect	Classification	
		Major	Minor
Workmanship and assembly (general) Continued	Evidence of black or white spots on canteen.	119 * 1/	208* 1/
	Any component misplaced or misaligned.	116	
	Any gate not trimmed flush with the molded surface.		209
	Any component missing.	117	
	Any component loose or not attached as specified. Heat seal of neck to canteen not complete or not free of excess material.	118 * 1/ 210 119 * 1/ 211	
	Flash at the bottom pinch-off area of the bladder parting line extending more than 1/32 inch.		212
	Flash appearing on parting lines except as above.		213
Workmanship and assembly of cap.	Canteen cap will not screw onto neck of canteen	118	
	Cap binds when screwing onto neck of canteen	119 * 1/ 212	
Workmanship of threads	Threads on canteen neck stripped, not permitting tight closure.	120	
	Threads not smooth.	121	
	Threads not fully formed, not free from flash, misaligned at parting line or other imperfection.	122	
Workmanship and assembly of neck	Sealing surface of lip is not flat, smooth, and free from sink marks or depression.	123 * 1/ 213	
	Flash appearing on parting line.	124 * 1/ 214	
	Neck not threaded with buttress type threads for a minimum of two full turns.	125 * 1/ 215	
	Strap retention groove, malformed.	126	
	Strap retention groove has rough or feathered ledge.	127 * 1/ 216	
	Rough surface below lower ledge of retention groove.		217

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TABLE IV. Defects. – Continued

Examine	Defect	Classification	
		Major	Minor
Workmanship and assembly of neck – (Continued.)	Void or rough area in interior of neck.	128 * 1/	218
Cleanliness.	Dirt, grease, dust or other foreign matter inside of the canteen.	129 * 1/	219
Marking	Missing, incomplete, illegible, misspelled, not in a specified location or not applied in a specified manner.		220
	Marking impairs the strength of the wall to which it is applied.	130 * 1/	221

1/ *Defects with an asterisk in the classification column are to be classified as follows:
Major defect when seriously affecting serviceability or appearance.
Minor defect when affecting serviceability or appearance, but not seriously.

4.4 Examination for wall thickness. The end item shall be examined for wall thickness requirements specified in 3.3.1. For this examination, the bladder of the canteen shall be cut in half along the parting line and each half cut 90^0 to the parting line to give four quarters. Measurements shall be made on each quarter selected. Five (5) measurements shall be made along with the parting line and five (5) areas selected throughout the body of each quarter. A failure of any sample unit to meet the requirement shall be scored as a defect. Measurements on a sample unit shall not be averaged. The sample unit shall be one (1) finished canteen, and the sample size shall be five (5) regardless of the size. Any defect found in this examination shall cause the lot represented by the sample to be rejected. Measurements shall be to the nearest 0.001 inch.

4.4.1 Dimensional examination of canteen. Any dimension, other than wall thickness that is not within the specified tolerances, shall be classified as a defect.

4.5 End item testing. Testing of the completely fabricated canteen shall be performed in accordance with Table V for the characteristics shown therein. The sample unit shall be four (4) completely fabricated canteens. The sample size for the designated lot size shall be as shown below. Any test failure shall be cause for rejection of the lot. All test reports shall content the individual values utilized in expressing the final result.

Lot size (canteens)	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 or more	5

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TABLE V. End item/material testing.

Characteristic	Requirement	Test method	Number of determinations per sample unit	Results reported as	
				Pass or fail	Numerically to nearest
Capacity	3.3	4.3.2	1	X	
Weight	3.3.1	1/	1	-	0.5 g
Porosity	3.4.1	4.3.1	1	X	
Leak resistance	3.4.2	4.3.2	1	X	
Drop resistance	3.4.3	4.3.3	1	X	

1/ A balance, sensitive to 0.5. gram, shall be used.

4.6 Toxicity test. When the toxicity requirement (see 3.6) can be demonstrated with historical use data on the finishing treatments used, toxicity testing may not be required (see 6.2)

5. PACKAGING**

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES:

6.1 Intended use. The collapsible canteen with or without a cover is intended for use in temperate and tropical environments by individual military personnel for carrying drinking water.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. The specific issue of individual documents referenced (see 2.).
c. When first article is required (see 3.1).
d. When toxicity is required (see 3.6).
e. Inspection conditions (see 4.5).
g. Performance inspection with acceptance quality limit (AQL) (see 4.3.2).
h. Packaging (see 5).

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6.3 First article. When a first article inspection is required, it will be inspected and approved under the appropriate provisions of FAR 52.209-4. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Certificate of compliance. The contracting activity may select to accept a certificate of compliance for that specified requirement.

6.5 Visual shade matching. In 2017, Option A of AATCC Evaluation Procedure 9, Visual Assessment of Color Difference of Textiles was changed to Option C. NOTE: In case of confusion, the viewing geometry should be "The specimen plane and illumination source will be parallel to each other and aligned so that the light flux is incident at the center of the specimen plane, which is set at a 35 (± 5°) angle relative to the horizontal. The observer will view the specimens at a 90° angle, relative to the plane of the specimens".

6.5.1 In 2022, the U.S. Military (All Services) accepted D65 as the new preferred light source for visual shade matching due to the supply chain issues with D75. DD1222's will reflect which light source is used. If the light source is not indicated on the DD1222, D65 will be assumed by the Government Shade Laboratory.

6.6 Subject term (key word) listing.

Container
Cover
Drinking
Hydration
Strap

Custodians:
Army – GL
Navy – NU

Preparing activity:
DLA-CT

(Project: 8465-2026-001)

Review activities:
Navy-MC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using ASSIST at https://assist.dla.mil.

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